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Roll No. ....

BE II (ME)

Kine. & Dyna. of Mach.

SECOND B.E EXAMINATION - 2023

MECHANICAL ENGINEERING

(IV SEMESTER CBCS)

ME - 253 : Kinematics & Dynamics of Machines

(M)

Time - Three Hours

Maximum Marks - 80

Note:- (i) Attempt any **FIVE** questions.

(ii) Q. No. 1 complete solution (including calculation) is to be given in the drawing sheet.

(iii) All questions carry equal marks.

(iv) Assume suitable data, if not given and mention it clearly.

(v) Draw neat sketches, wherever necessary.

1. A flat-faced mushroom follower is operated by a uniformly rotating cam. The follower is raised through a distance of 35mm in 120° rotation of the cam, remains at rest for the next 30° and is lowered during further 90° rotation of the cam. The raising of the follower takes place with cycloidal motion and the lowering with uniform acceleration and deceleration. However, the uniform acceleration is 2/3of the uniform deceleration. The least radius of the cam is 25 mm which rotates at 450 rpm. Draw the cam profile and determine the values of the maximum velocity and maximum acceleration during rising, and maximum velocity and uniform acceleration and deceleration during lowering of the follower. 16

2. (a) In a Hartnell governor, the radius of rotation of the balls is 50 mm at the minimum speed of 280 rpm. The length of the ball arm is 130 mm and the sleeve arm is 80 mm. The mass of each ball is 3 kg and the sleeve is 5 kg. The stiffness of the spring is 25 N/mm. Determine the (i) speed when the sleeve is lifted by 55 mm (ii) initial compression of the spring (iii) governor effort (iv) power. 8

(b) What is the controlling force of a governor? How are the controlling force curves drawn? How do they indicate the stability or instability of a governor? Indicate the shape of such a curve for an isochronous governor. 8

3. (a) A band and block brake having 12 blocks, each of which subtends an angle of 18° at the centre, is applied to a rotating drum with a diameter of 600 mm. The blocks are 75 mm thick. The drum and the flywheel mounted on the same shaft have mass of 1900 kg and have a combined radius of gyration of 600 mm. The two ends of the band are attached to pins on the opposite sides of the brake fulcrum at distances of 42 mm and 158 mm from it. If a force of 300 N is applied on the lever at a distance of 950 mm from the fulcrum, find the (i) maximum braking torque (ii) angular retardation of the drum (iii) time taken by the system to the stationary from the rated speed of 350 rpm.

Take coefficient of friction between the blocks and the drum as 0.32. 8

(b) Describe with the help of a neat sketch the principles of operation of an internal expanding shoe. Derive the expression for the braking torque. 8

4. (a) Describe the construction and operation of a prony brake or rope brake absorption dynamometer with neat sketch. 8

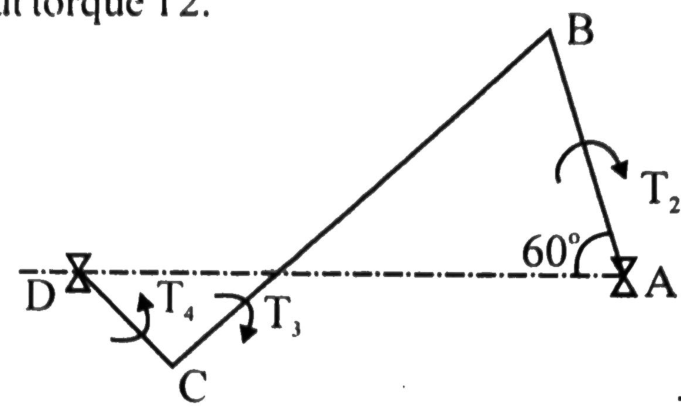
(b) A torsion dynamometer is fitted on a turbine shaft to measure the angle of twist. It is noticed that the shaft twist 2.5° in a length of 5 meters at 750 rpm. The shaft is solid and has a diameter of 250 mm. If the modulus of rigidity for the shaft is 82 GPa, find the power transmitted by the machine. 8

5. (a) A four - wheeled trolley car has a total mass of 3500 kg. Each axle with its two wheels and gears has a total moment of inertia of 38 kg.m^2. Each wheel is of 450 mm radius. The centre distance between two wheels on an axle is 1.6m. Each axle is driven by a motor with a speed ratio of 1:3. Each motor along with its gear has a moment of inertia of 19 kg.m^2 and rotates in the opposite direction to that the axle. The centre of mass of the car is 1.2 m above the rails. Calculate the limiting speed of the car when it has to travel around a curve of 250 m radius without the wheels leaving the rails. 8

(b) Explain the gyroscopic effect on ships. Also explain the axis of spin, precession, and gyroscopic couple with neat sketches.

C = f
12 = f.
= ⊖.

6. (a) In a four-link mechanism shown in Figure 1, torque T3 and T4 have magnitude of 35 N.m and 25 N.m, respectively. The link lengths are AD = 850 mm, AB = 350 mm, BC = 750 mm and CD = 450 mm. For the static equilibrium of the mechanism, determine the required input torque T2.



(b) Define and explain the superposition theorem as applicable to a system of forces acting on a mechanism.

8

7. Write short notes (any four): 4x4=16

(a) Describe the function of a simple Watt governor. What are its limitations?

(b) Explain the tractive resistance in wheeled vehicle.

(c) What is the principle of virtual work? Explain.

(d) Explain the gyroscopic effect on four-wheeled vehicles.

(e) Discuss various types of cams and draw neat sketches.

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BE II (ME)

3 **FM - I**

SECOND B.E. EXAMINATION - 2023
MECHANICAL ENGINEERING
(IV SEMESTER CBCS)
ME 255 A : Fluid Mechanics - I (M)

Time - Three Hours

Maximum Marks - 80

Note:- (i) Attempt any **FIVE** questions.

(ii) Answer should be to the point.

(iii) All questions carry equal marks.

(iv) Whatever data are missing, assume and mention them clearly.

1.(a) (i) Explain the effect of temperature on viscosity of liquids and gases. 4

(ii) Define Newtonian and Bingham fluid graphically. Give at least two examples for each of them. 4

1000×9.81×0.8 − 0.8×...

− 0.9×1000×9.81×0.5 = P_B

(b) Two horizontal flat plates are placed 0.15 mm apart and the space between them is filled with an oil of viscosity 1 poise. The upper plate of area 1.5 m² is required to move with a speed of 0.5 m/s relative to the lower plate. Calculate the necessary force and power required to maintain this speed. 8

2.(a) Derive an expression for hydrostatic force acting on a submerged plane surface inclined with respect to free surface of liquid. 8

(b) Determine the difference of pressure between pipes A and B when connected to an inverted U-tube differential manometer containing oil of specific gravity 0.8 as the manometric liquid. The pipe A conveys water and a fluid of specific gravity 0.9 flows through the pipe B. The position of manometric liquid in the manometer limbs is indicated in figure - 1

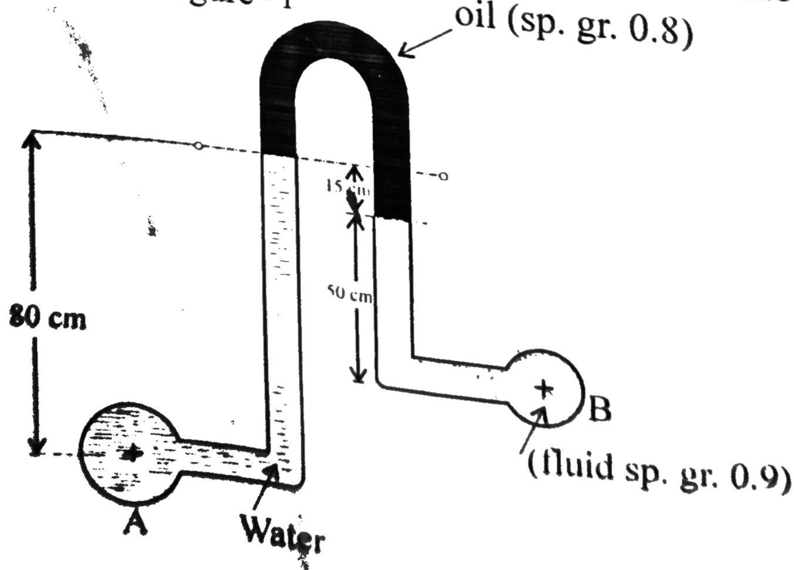

Figure - 1

P_A − 1000×9.8×0.8 − 0.8×1000×9.8×0.15

− 0.9×1000×9.8×0.50

= P_B

If P_B (gauge) = 5×10^4 N/m² and the barometer reading is 730 mm of mercury, find the pressure in pipe A in meters of water absolute. 8

3.(a) What do you understand by metacenter of a floating body? Define metacentric height. What are the conditions of equilibrium of floating bodies? 8

(b) A wooden block of width 15cm x depth 30 cm and length 150 cm has a mass of 54 kg. Can the block float with 30 cm side vertical in water? Comment on the stability of the block. 8

4.(a) Derive continuity equation in three dimensions for Cartesian coordinates for general flow. Simplify it for various cases of flow. 8

(b) The velocity component in a two dimensional flow field for an incompressible fluid are expressed as

$$u = \frac{y^3}{3} + 2x - x^2 y$$

$$v = xy^2 - 2y - \frac{x^3}{3}$$

(i) Show that these functions represent a possible flow.

(ii) Show that these functions represent an irrotational flow.

(iii) Obtain an expression for stream function ψ.

(iv) Obtain an expression for velocity potential ϕ. 8

5.(a) Derive Euler's equation of motion along a stream line for an ideal fluid stating clearly all the assumptions. 8

(b) Prove that discharge through a right angled triangular V-notch is given by

$$\theta = \frac{8}{15} C_d \sqrt{2g}\, H^{5/2}$$

Where symbols have their usual meanings. 8

6.(a) What is the principle of Venturimeter? Explain. Derive an expression for discharge through a venturimeter. Why actual discharge is less than theoretical discharge?

8

(b) Consider force F on propeller of an aircraft. It depends on forward speed of aircraft U, density of air ρ, viscosity of air μ, diameter of propeller D and speed of rotation of propeller N. Show by dimensional analysis using Buckingham's π Theorem:

$$\frac{F}{\rho U^2 D^2} = f\left[\frac{UD\rho}{\mu}, \frac{ND}{U}\right]$$ 8

7. Write short notes on the following : 4x4=16

(a) Stream line, path line and streak line.

(b) Measurement of static, stagnation and dynamic pressure.

(c) Froude Number and Weber number & their applications.

(d) Vorticity and Circulation.

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BE II (ME)

3 **MD - I**

SECOND B.E. EXAMINATION - 2023

MECHANICAL ENGINEERING

(IV SEMESTER CBCS)

ME 254 A : Machine Design - I (M)

Time - Three Hours

Maximum Marks - 80

Note:- (i) Attempt any **FIVE** questions.

(ii) Answer should be to the point.

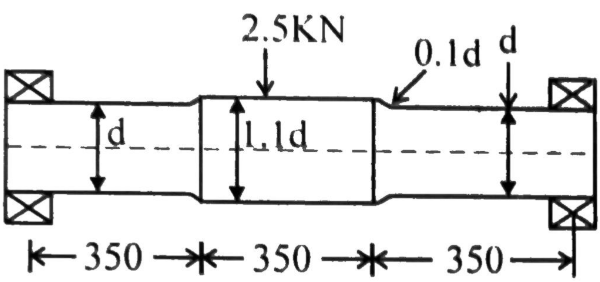
(iii) All questions carry equal marks.

1.(a) Define machine design and morphological analysis used in Designing Process. 8

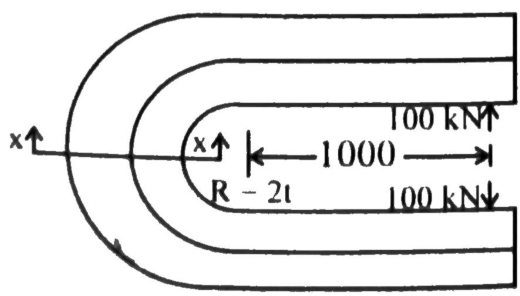
(b) Describe the designers responsibility and various aspects of design used in Machine design. 8

2.(a) Why we required different theories of failure in machine design? Explain different theories of failure. 8

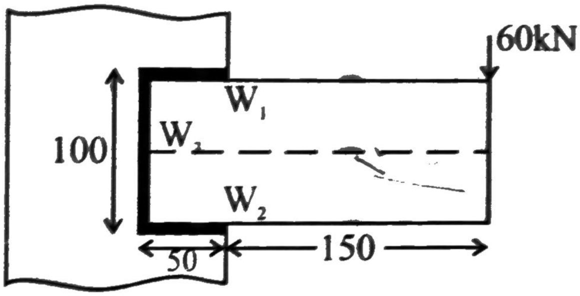
(b) A non - rotating shaft supporting a load of 2.5 kN as shown in Fig. The shaft is made of brittle material, with an ultimate tensile strength of 300 N/mm². The factor of safety is 3. Determine the dimensions of the shaft. 8



3. The C - frame of a 100 kN capacity press is shown in Fig. The material of the frame is grey cast iron FG 200 and the factor of safety is 3. Determine the dimensions of the frame. 16



4 A welded connection as shown in fig. is subjected to an eccentric force of 60 kN in the plane of the welds. Determine the size of the welds, if the permissible shear stress for the weld is 100 N/mm². Assume static conditions. 16



5. A safety valve, 50 mm in diameter is to blow off at a pressure of 1.5 NPa. It is held on its seat by means of a helical compression spring with an initial compression of 25 mm. The maximum lift of the valve is 10 mm. The spring index can be taken as 6. The spring is made of patented and cold-drawn steel wire with ultimate tensile strength of 1500 N/mm² and modulus of rigidity of 81370 N/mm². The permissible shear stress for the spring wire should be taken as 30% of the ultimate tensile strength. Design the spring and calculate.

(i) Wire diameter

(ii) Mean coil diameter

(iii) No. of active turns

(iv) Total no. of turns

(v) Solid length

(vi) Free length

(vii) Pitch at the coil 16

Design a muff coupling to connect two steel shafts transmitting 25 kW power at 360 rpm. The shafts and key are made of plain carbon steel 30 C8 ($S_{yt} = S_{yc} = 400$ N/mm^2). The sleeve is made of grey cast iron FG200 (S_{ut}= 200 N/mm^2). The factor of safety for the shaft and key is 4. For the sleeve the factor of safety is 6 based on ultimate strength. 16

A right angled bell- crank lever is to be designed to raise a load of 5 kN at the short arm end. The lengths of short and long arm are 100 and 450 mm respectively. The lever and the pins are made of steel 30 C8 (S_{yt} = 400 N/mm^2) and the factor of safety is 5. The permissible bearing pressure on the pin is 10 N/mm^2. The lever has a rectangular corss-section and the ratio of width to thickness is 3:1. The length to diameter ratio of the fulcrum pin is 1.25:1. Calculate.

(i) The diameter and the length of the fulcrum pin.

(ii) The shear stress in the pin.

(iii) The dimensions of the boss of the lever at the fulcrum.

(iv) The dimensions fo the cross-section of the lever.

Assume that the arm of the bending moment on the lever extends up to the axis of the fulcurm. 16

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BE II (ME)

F&WE

SECOND B.E. EXAMINATION, 2023
MECHANICAL ENGINEERING
(IV SEMESTER) (CBCS)
ME 252 A : Foundry & Welding Engineering(M)

Time – Three Hours

Maximum Marks – 80

Note: (i) Attempt any **Five** questions.

(ii) Answer should be to the point.

(iii) All questions carry equal marks.

1. (a) Consider a foundry for steel castings. What types of pattern would you suggest for small scale, medium scale and large scale production? Also Suggest various Pattern allowances for them. 10

(b) What are the various test methods for moulding sand to check the quality of sand? Describe any four methods. 6

2. (a) Explain in detail how investment castings are produced with suitable diagrams. 10

(b) Compare hot chamber with cold chamber die casting mentioning various advantages and disadvantages. 6

3. (a) What are the various casting defects? How can we overcome or reduce them? 6

(b) Discuss advantages. limitations and applications

(i) Carbon dioxide moulding
(ii) Shell moulding
(iii)Centrifugal casting
(iv) Continuous casting 10

4. (a) How metal transfer in welding occurs? Explain metal transfer methods with suitable diagrams. 10

(b) Compare TIG and MIG welding also discuss their applications. 6

5. (a) What are various flame types in oxy acetylene gas welding show with proper diagram and various zones? 8

(b) Explain various welding joints and welding positions with suitable diagrams. 8

6. (a) Attempt any four, write short notes:- 4x4

(i) Soldering and Brazing
(ii) Gate, runner, riser
(iii)Cupola furnace
(iv)Friction welding
(v) Thermal welding
(vi)AC & DC welding and their applications

7. (a) What are various resistance welding process. Discuss briefly with diagram. 10

(b) Non destructive testing used in both casting and welding. Explain any three methods. 6

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Roll No. ...11.2.1.2.............

BE II (ME)

3 RAC

SECOND B.E. EXAMINATION - 2023

MECHANICAL ENGINEERING

(IV SEMESTER CBCS)

ME 251 A : Refrigeration & Air Conditioning (M)

Time - Three Hours

Maximum Marks - 80

Note:- (i) Attempt any **FIVE** questions.

(ii) Answer should be to the point.

(iii) All questions carry equal marks.

Use of Psychometric charts and Refrigeration table and P-h charts and steam table are permitted.

1. (a) Derive the expression for COP of a reversed Carnot refrigeration cycle. What are the limitations of the cycle? 8

(b). Describe boot-strap cycle of air refrigration system, with a schematic diagram and show the cycle on T-S diagram. 8

2.(a) Explain the effect of superheating in compressor and under cooling in vapour compression system. Show these on T-S and P-h diagram. 8

(b) A refrigeration machine using R-12 as refrigrant operates between the pressure 2.5 bar and 9 bar. The compression is isentropic and there is no undercooling in the condenser.

The vapour is in dry saturated condition at the beginning of the compression. Estimate the theoretical coefficient of performance. If the actual COP is 0.65 of the theoretical value, calculate the net cooling produced per hour. The refrigerant flow is 5 kg per minute.

Properties of refrigerant are: 8

Pressure (bar)	Saturation temp' °C	Enthalpy (KJ/kg)		Entropy of Saturated Vapour (KJ KgK)
		Liquid	Vapour	
9.0	36	70.55	201.8	0.6836
2.5	-7	29.62	184.5	0.7001

3 (a). Derive an expression for the COP of an ideal vapour absorption system is terms of temperature T_G at which heat is supplied to the generator, the temp' T_E at which

heat is absorbed in the evaporator and the temp' T_c at which heat is discharged from the condenser and absorber. 8

(b) Draw a neat diagram of lithium bromide water absorption system and explain its working. 8

4. (a) What do you mean by ODP and GWP related to refrigerants? Discuss the physical and thermodynamic properties of the ideal refrigerant. 8

(b) A simple air cooled system is used for an aeroplane having a load of 10 tonnes. The atmospheric pressure and temp' are 0.9 bar and 10°C respectively. The pressure increases to 1.013 bar due to ramming. The temp' of air is reduced by 50°C in the heat exchanger. The pressure in the cabin is 1.01 bar and the temp' of air leaving the cabin is 25°C. Determine :

(i) Power required to take the load of cooling in the cabin

(ii) COP of the system

Assume that all the expansion and compressions are isentropic. The pressure of the compressed air is 3.5 bar. 8

(Contd.)

an

5. (a) Establish the following expressions for air vapour mixture:

Specific humidity $w = 0.622 \times \frac{P_v}{P_b - P_v}$

P_v = Partial Pr. of water vapour

P_b = Barometric pressure

4

(b) For a sample of air having 22°C DBT, relative humidity 30% at barometric pressure of 760 mm of Hg, Calculate

(i) Vapour Pressure

(ii) Humidity ratio

(iii) Enthalpy

6

(c) At a certain locality, the dry bulb temp' of air is 30°C and the relative humidity is 40%. Determine the specific humidity and dew point and wet bulb temp' of air. If the air is cooled in an air washer using recirculated spray water and having a humidifying effeciency of 0.9. What are dry bulb temp' and dew point temp' of air leaving the air washer?

6

6. (a) Explain with a neat sketch the working of Year - Round Air Conditiong System.

8

(b) What is "effective temperature"? What factors affect effective temp'?

8

. Write short notes on any four

(i) Dry air rated temp^r (DART)

(ii) Automatic expansion value

(iii) Electrolux refrigerator

(iv) Comfort charts

(v) Chemical dehumidification

4x4 = 16

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Roll No. 22U1AE(1122

BE II (CE)

2 **Open Elective - I**

SECOND B.E. EXAMINATION - 2023

CIVIL ENGINEERING

(IV SEMESTER CBCS)

CE 291 A : Open Elective - I :

Energy Efficient Building Design

Time - Three Hours

Maximum Marks - 100

Note:- (i) Attempt any **Five** questions.

(ii) Answer should be to the point.

(iii) All questions carry equal marks.

1. (a) What is the Integrated approach to the built environment? 10

(b) What are the different types of external environment that affect the built environment? Explain with examples. 10

2. (a) Describe the classification of climates and their effects on the built environment. 10

(b) What is site climate, and how does it affect the design of buildings? 10

3. (a) Explain the factors that affect thermal comfort in indoor environments. 10

(b) What are the different comfort indices used to evaluate indoor thermal comfort? 10

4. (a) Explain the concept of heat gains due to solar radiation in buildings. What are the factors affecting heat gain, and how can they be controlled? 10

(b) What are the means of thermal control in buildings? Explain the mechanical and structural methods to control thermal conditions in buildings. 10

5. (a) Explain the concept of day - lighting and its benefits in building design. 10

(b) What is vibration and how does it affect the built environment? Explain the different methods of vibration control. 10

6. (a) What are the sources of noise in the built Environment? How can they be controlled. 10

(b) What are the factors to consider while selecting mechanical and electrical services for a building? How can these services be optimized for energy efficiency and cost - effectiveness? 10

7. Write short notes on the following (Any Four) 5X4=20

(a) External environment and Built environment

(b) Micro - climate and Site climate

(c) Types of lifts commonly used in buildings

(d) Heat loss from a building

(e) Shape of buildings for thermal comfort